

SURVEILLANCE BOT – MOBILE CONTROLLED SURVEILLANCE ROBOT USING ESP32

A PROJECT REPORT

Submitted by

NITISH S (412725104073)

THIRUMURUGAN D (412725104108)

SIDDHARTH P (412725104103)

MUGILAN L (412725104066)

PRAVEEN KUMAR P (412725104079)

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ANNA UNIVERSITY: CHENNAI 600 025**

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BONAFIDE CERTIFICATE

Certified that this project report “ **SURVEILLANCE BOT – MOBILE CONTROLLED SURVEILLANCE ROBOT**” is the bonafide work of “ **NITISH S (412725104073), THIRUMURUGAN D (412725104108), SIDDHARTH P(412725104103), MUGILANL (412725104066), PRAVEEN KUMAR P (412725104079)** ” who carried out this project under my supervision.

SIGNATURE

DR. R. PORSELVI, M.E., Ph.D.,
HEAD OF THE DEPARTMENT
ASSOCIATE PROFESSOR
Department of ECE
Tagore Engineering College
Rathinamangalam,
Chennai-127

SIGNATURE

Mrs. R. HEMA, M.Tech.,
SUPERVISOR
ASSISTANT PROFESSOR
Department of ECE
Tagore Engineering College
Rathinamangalam,
Chennai-127

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ABSTRACT

In today's world, robotics technology has become increasingly important in reducing human effort and improving safety in dangerous environments. Many situations such as disaster zones, military operations, and hazardous industrial areas require monitoring where human presence may be risky or difficult. To address this problem, this project focuses on developing a mobile-controlled surveillance robot called the Surveillance Bot that can perform movement operations using simple wireless commands from a smartphone. The main idea of this system is to control a robot remotely through wireless communication. A mobile application such as Dabble is used to send commands like "forward," "backward," "left," and "right."

These commands are received through Bluetooth by the ESP32 microcontroller, which acts as the brain of the system. Based on the programmed instructions, the microcontroller generates control signals to operate the motors. The robot is built using four DC motors connected to a four-wheel drive robot chassis. A motor driver module is used to control the motors because the ESP32 cannot directly supply enough current to run them. Depending on the command given by the user, the motor driver controls the direction and speed of the motors, allowing the robot to move smoothly in different directions. The entire system is powered by a battery, making the robot portable and easy to operate. One of the main advantages of this project is that it uses wireless technology to control a real-world device remotely in a simple and effective way. The system is cost-effective and built using easily available components, making it suitable for educational purposes and practical applications. This robot can be used for surveillance and monitoring in places where human access is dangerous, helping to reduce risk and improve safety.

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LIST OF ABBREVIATION

ABBREVIATION	DEFINITION
ESP32 DEV	Microcontroller Module
USB	Universal Serial Bus
DC	Direct Current
GPIO	General Purpose Input Output
GND	Ground
TX	Transmitter
RX	Receiver
VCC	Voltage Common Collector

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Robotics technology has become an essential part of modern engineering and automation, aiming to reduce human effort and improve safety in various environments. In many situations such as disaster zones, military operations, industrial plants, and hazardous areas, human presence may be risky or even impossible. Monitoring such environments manually can expose individuals to dangerous conditions, including fire, toxic gases, structural collapse, or extreme temperatures. Therefore, it is necessary to develop innovative robotic systems that can perform surveillance and monitoring tasks remotely while ensuring human safety.

In recent years, advancements in embedded systems, wireless communication, and robotics have enabled the development of mobile robots capable of performing various operations with minimal human intervention. These robots are widely used for inspection, surveillance, and data collection in places where direct human access is limited. Many existing robotic systems use wired control mechanisms or complex remote-control devices to operate. While these methods can be effective, they often limit mobility and flexibility due to the restrictions of physical connections. Additionally, wired systems may not be suitable for long-distance control or emergency situations where quick deployment is required.

To overcome these limitations, wireless communication technologies such as Bluetooth and Wi-Fi provide a practical and efficient solution for controlling robots remotely. Wireless control allows users to operate robots from a safe distance using simple devices like smartphones. This approach enhances flexibility, improves response time, and reduces the need for specialized control equipment. Among various wireless technologies, Bluetooth communication is

widely used due to its low power consumption, reliability, and ease of integration with mobile devices.

The mobile-controlled surveillance robot, known as the **Surveillance Bot**, is designed to perform movement operations using wireless commands from a smartphone. The system is built around the ESP32 microcontroller, which acts as the central processing unit of the robot. The ESP32 is selected because of its built-in Bluetooth and Wi-Fi capabilities, high processing speed, and energy efficiency. The robot consists of a four-wheel drive mechanism powered by DC motors mounted on a sturdy robot chassis. A motor driver module is used to control the motors, as the microcontroller alone cannot provide sufficient current to operate them directly.

The working principle of the Surveillance Bot is based on wireless communication between a mobile device and the robot. A mobile application, such as Dabble, is used to send control commands like “forward,” “backward,” “left,” and “right.” These commands are transmitted via Bluetooth to the ESP32 microcontroller, which processes the input signals according to the programmed logic. The microcontroller then sends control signals to the motor driver module, which regulates the direction and speed of the motors. As a result, the robot moves smoothly in the desired direction based on user input.

One of the key advantages of this robotic system is its simplicity and ease of operation. Unlike traditional surveillance systems that require complex equipment or manual control, the mobile-controlled robot can be operated using a smartphone interface, making it convenient and user-friendly. This significantly reduces the learning effort required to use the system and allows quick deployment in emergency situations. Furthermore, the use of commonly available components such as microcontrollers, motor drivers, and DC motors helps in reducing the overall cost of the system, making it affordable for educational and practical applications.

Another important aspect of the Surveillance Bot is its potential application in

surveillance and monitoring. The robot can be used to observe environments where human entry may be unsafe, such as fire-affected buildings, disaster zones, military areas, or industrial sites with hazardous materials. By acting as a remotely operated system, the robot helps reduce the risk to human life while providing real-time monitoring capabilities. The design can also be further enhanced by adding features such as cameras, sensors, obstacle detection systems, or Internet of Things (IoT) connectivity for improved functionality and performance.

The development of mobile-controlled robots also plays a significant role in education and research. Building such systems helps students understand the fundamentals of robotics, embedded systems, motor control, and wireless communication. It provides practical experience in designing and programming real-world devices, which is essential for developing technical skills in modern engineering fields. Additionally, these robots can be used as training tools in robotics laboratories and engineering projects to demonstrate automation and remote-control technologies.

Despite the advantages, certain challenges remain in the implementation of mobile-controlled robotic systems. Factors such as signal interference, battery life, response time, and mechanical stability must be carefully considered during design and development. However, with continuous advancements in wireless communication and microcontroller technology, these limitations can be minimized, leading to more reliable and efficient robotic solutions.

In conclusion, the **Surveillance Bot** represents a practical and effective approach to developing a mobile-controlled surveillance robot. By combining robotics with wireless communication, the system provides a safe and efficient method for monitoring hazardous environments remotely. Its simplicity, cost-effectiveness, and adaptability make it a valuable project for educational purposes and real-world applications, with significant potential for future improvements and technological advancements.

IMPROVEMENTS FROM LITERATURE SURVEY**2.1 LITERATURE SURVEY**

Literature survey not only reviews what research has been conducted in the past on the topic, but it also evaluates, summarizes, compares, and contrasts various scholarly articles, research papers, and technical studies that are directly related to the current project. The purpose of this literature survey is to understand existing robotic surveillance systems, wireless-controlled robots, and embedded system technologies that contribute to the development of the Surveillance Bot.

1.Design and Implementation of Bluetooth Controlled Surveillance Robot

Author: R. Kumar, S. Prakash, A. Sharma

YEAR:2022

In modern robotics, remote-controlled surveillance robots are widely used in hazardous environments such as military operations, disaster zones, and industrial inspection areas. This study presents the design and implementation of a Bluetooth-controlled surveillance robot capable of performing basic monitoring and movement operations using a smartphone application.

The robot is built using a microcontroller, DC motors, a motor driver module, and a Bluetooth communication system. The main objective of the project was to create a simple and reliable robotic platform that can be remotely controlled without direct human presence in dangerous locations. The system operates through wireless communication between the mobile device and the robot, allowing the user to send movement commands such as forward, backward, left, and right.

The research demonstrated that Bluetooth-based communication provides an efficient and low-cost method for controlling robotic systems over short distances. The robot was successfully tested in indoor environments, showing stable

performance and quick response to commands. However, the system had limitations in terms of obstacle detection and environmental awareness.

This study highlights the importance of wireless communication in robotics and provides a strong foundation for developing mobile-controlled surveillance robots. It also suggests that future systems can be improved by integrating sensors and cameras for better monitoring capabilities.

2. Development of Wi-Fi Based Mobile Surveillance Robot Using ESP32

Authors: P. Ramesh, V. Karthik, N. Suresh

Year: 2023

The increasing demand for remote monitoring systems has led to the development of Wi-Fi-based surveillance robots capable of transmitting real-time data over long distances. This research focuses on designing a surveillance robot using the ESP32 microcontroller due to its built-in Wi-Fi and Bluetooth capabilities, high processing speed, and energy efficiency.

The system consists of an ESP32 module, motor driver, DC motors, and a camera module for live video streaming. The robot is controlled through a mobile application connected via Wi-Fi, enabling users to monitor and control the robot remotely from a greater distance compared to Bluetooth-based systems.

The working principle involves sending control signals from the smartphone to the ESP32 through wireless communication. The ESP32 processes the received commands and sends signals to the motor driver to control the robot's movement. Additionally, the camera captures real-time images and transmits them to the user for surveillance purposes.

Experimental results showed that the robot provided reliable communication and efficient performance in monitoring applications. However, the system required a stable internet connection for proper operation.

This study demonstrates the advantages of using ESP32 in robotic systems and highlights the potential of integrating wireless communication and video surveillance technologies in modern robotics.

3. Design of Obstacle Detection Robot Using Ultrasonic Sensor

Authors: M. Ahmed, K. Rahman

Year: 2021

Obstacle detection is an essential feature in autonomous robotic systems, especially in environments where navigation safety is critical. This study presents the design and implementation of a robot capable of detecting obstacles using an ultrasonic sensor and avoiding collisions automatically.

The robot uses an ultrasonic sensor to measure the distance between the robot and nearby objects. When an obstacle is detected within a predefined distance, the microcontroller automatically stops the robot or changes its direction to prevent collision. The system also includes DC motors, a motor driver module, and a power supply unit.

The main objective of the research was to improve the safety and reliability of robotic navigation systems. The experimental results showed that the robot successfully detected obstacles within a range of 2 to 400 centimetres with high accuracy.

Although the robot performed well in obstacle detection, it lacked remote control functionality and advanced surveillance features. The study emphasizes the importance of sensor integration in robotic systems and suggests combining obstacle detection with wireless control for enhanced performance.

This research provides valuable insights into improving the safety features of surveillance robots like the Surveillance Bot.

4. Mobile Controlled Robot Using Arduino and Bluetooth Technology

Authors: S. Gupta, R. Singh, A. Verma

Year: 2020

This study focuses on the design and development of a mobile-controlled robot using Arduino and Bluetooth communication technology. The robot is controlled through a smartphone application that sends commands to the Arduino microcontroller via a Bluetooth module.

The system consists of an Arduino board, motor driver, DC motors, Bluetooth module, and rechargeable battery. The main objective of the project was to create an affordable and easy-to-build robotic system suitable for educational and experimental purposes.

The robot operates by receiving control signals from the mobile application and converting them into motor control commands. The motor driver module controls the direction and speed of the motors, enabling the robot to move in different directions.

Testing results showed that the robot responded quickly to user commands and maintained stable performance during operation. However, the system had limited processing capability compared to modern microcontrollers like ESP32.

This research demonstrates the effectiveness of Bluetooth communication in controlling robotic systems and provides a basic framework for developing mobile-controlled robots.

5. Surveillance Robot for Disaster Management Applications

Authors: A. Patel, D. Shah

Year: 2022

Disaster management operations often require monitoring dangerous environments where human access is limited. This study presents the design of a surveillance robot specifically developed for disaster response and rescue missions.

The robot is equipped with sensors for temperature detection, gas leakage monitoring, and obstacle detection. It also includes a wireless communication system that allows remote operation and data transmission to a control centre.

The main objective of the research was to improve the safety of rescue workers by providing real-time environmental data from hazardous locations. The robot was successfully tested in simulated disaster conditions, demonstrating reliable performance and accurate sensor readings.

Despite its effectiveness, the system was relatively complex and expensive due to the integration of multiple sensors and communication modules.

This study highlights the practical applications of surveillance robots in emergency situations and emphasizes the importance of developing cost-effective robotic solutions for safety and monitoring purposes.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 OBJECTIVE

The main objective of this project is to design and develop a mobile-controlled surveillance robot called the **Surveillance Bot** using an ESP32 microcontroller. The system is intended to perform movement operations such as forward, backward, left, and right through wireless control using a smartphone. The robot is designed to provide a simple, reliable, and efficient method for remote monitoring and navigation in environments where human access may be dangerous or difficult. The design focuses on achieving smooth motor control, stable wireless communication, and real-time response while maintaining low cost and ease of implementation. The robot is built using commonly available components such as an ESP32 microcontroller, motor driver module, DC motors, and a rechargeable power supply. The overall objective is to enhance safety, reduce human effort, and demonstrate the practical application of robotics and embedded systems in surveillance and monitoring tasks

3.2 PROPOSED SYSTEM WORKING METHODOLOGY

Mobile Command Input and Wireless Communication

The system begins operation when the user sends movement commands using a smartphone application such as Dabble via Bluetooth communication. The mobile device acts as a remote controller, allowing the user to operate the robot wirelessly from a safe distance. The commands include basic directional instructions such as forward, backward, left, right, and stop. These commands are transmitted through Bluetooth signals to the ESP32 microcontroller. The built-in Bluetooth module in the ESP32 receives the signals and converts them into digital data. The wireless

communication ensures reliable and fast transmission of commands without the need for physical connections, making the system flexible and easy to control

Command Processing and Control Logic

The ESP32 microcontroller acts as the brain of the system. It continuously monitors incoming Bluetooth signals and processes the received commands using pre-programmed logic. Each command is mapped to a specific movement of the robot.

For example:

- The **forward command** activates all motors to rotate in the forward direction
- The **backward command** reverses the motor rotation
- The **left command** turns the robot to the left side
- The **right command** turns the robot to the right side
- The **stop command** halts all motor movement

The microcontroller sends control signals to the motor driver module through its GPIO pins. The control logic ensures that the robot responds immediately to user commands and moves smoothly without delay

Motor Driver Operation and Robot Movement

The motor driver module plays an important role in controlling the movement of the robot. Since the ESP32 cannot supply sufficient current to drive the motors directly, the motor driver acts as an interface between the microcontroller and the DC motors.

When the ESP32 sends control signals, the motor driver amplifies the current and distributes power to the motors. The four DC motors mounted on the robot chassis rotate according to the direction specified by the control signals. This enables the robot to move forward, backward, or turn in different directions.

The use of a four-wheel drive mechanism ensures stable movement and better traction on different surfaces. The motor driver also protects the system from overload conditions and ensures smooth motor operation.

Power Supply and System Stability

Power management is an important aspect of the proposed system. The entire robot is powered by a rechargeable battery, which provides electrical energy to the ESP32 microcontroller, motor driver module, and DC motors.

A voltage regulator is used to maintain a stable power supply and protect the electronic components from voltage fluctuations. A common ground connection is maintained throughout the system to ensure proper signal transmission and prevent unexpected behaviour.

Safety features such as proper wiring, current protection, and secure component mounting help ensure reliable and safe operation of the robot during movement.

Real-Time Operation and Remote Control

The system operates continuously in real time, allowing the user to control the robot instantly through the mobile application. The ESP32 continuously listens for incoming commands and executes them immediately.

The wireless control system allows the robot to be operated from a safe distance, making it suitable for surveillance and monitoring applications. The robot can be used in areas such as:

- Disaster-affected zones
- Industrial environments
- Military surveillance areas
- Hazardous locations
- Remote monitoring situations

The system does not require complex setup or internet connectivity, making it reliable and easy to operate in different environments.

Flexibility and Future Expansion

The proposed system is designed with flexibility and scalability in mind. Additional components such as sensors, cameras, or GPS modules can be integrated into the robot without changing the basic system structure.

Future improvements may include:

- Camera module for live video monitoring
- Obstacle detection sensors
- Wi-Fi control instead of Bluetooth
- Autonomous navigation system

- Internet of Things (IoT) integration

These enhancements can increase the functionality and performance of the surveillance robot in real-world applications.

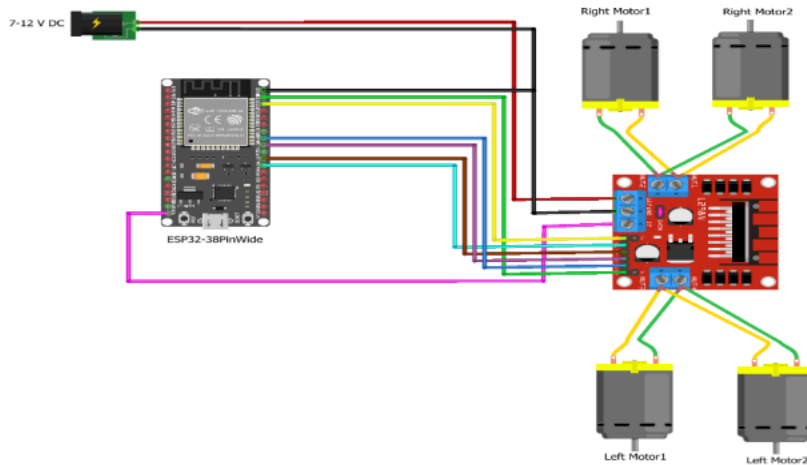


Figure 3.2 Proposed System Architecture

3.3 EXISTING SYSTEM

Existing surveillance robots are often controlled using wired connections, manual switches, or complex remote-control systems. Some advanced systems use cameras, sensors, and automated navigation technology. While these systems provide high performance, they are usually expensive and require advanced technical knowledge to operate.

Many traditional monitoring methods depend on human presence in dangerous environments, which increases the risk of accidents and injuries. In some cases, surveillance is carried out manually, making it time-consuming and unsafe. Additionally, complex robotic systems may require continuous maintenance and specialized equipment, which increases the overall cost.

These limitations highlight the need for a simple, cost-effective, and user-friendly surveillance system that can be controlled remotely using wireless technology.

Disadvantages of Existing System

The existing surveillance systems have several limitations that reduce their effectiveness and practicality. Many systems rely on manual monitoring, which exposes humans to dangerous environments and increases the risk of accidents. Some advanced robotic systems are expensive and require complex hardware and software for operation.

In addition, wired control systems restrict the movement range of the robot and reduce flexibility. Maintenance and installation costs can also be high, making such systems difficult to implement in small-scale applications. Furthermore, complicated user interfaces may require technical training to operate effectively.

These drawbacks emphasize the importance of developing a simple, affordable, and wireless-controlled surveillance robot like the Surveillance Bot.

CHAPTER 4

HARDWARE DESCRIPTION

4.1 ESP32 MICROCONTROLLER

The ESP32 microcontroller is a low-cost, high-performance microcontroller widely used in embedded systems, robotics, and Internet of Things (IoT) applications. It features integrated Bluetooth and Wi-Fi connectivity, making it suitable for real-time control and wireless communication tasks. In the proposed Surveillance Bot mobile-controlled surveillance robot, the ESP32 acts as the central control unit, processing commands received from the mobile application and generating control signals for the motor driver and DC motors..

4.1.1 Overview of ESP32 Microcontroller

The ESP32 is built around a dual-core Tensilica LX6 processor operating at speeds up to 240 MHz, providing high computational performance for embedded applications. It includes onboard memory for program storage and RAM for data processing, enabling efficient execution of control programs in real time.

The ESP32 microcontroller provides multiple General-Purpose Input/Output (GPIO) pins that can be used for digital input/output, analogy input, and Pulse Width Modulation (PWM) signals. These features allow easy interfacing with components such as motor drivers, sensors, and communication modules used in the surveillance robot system.

The ESP32 also supports wireless communication through built-in Bluetooth and Wi-Fi modules. In this project, the Bluetooth feature is used to receive commands from a smartphone application, allowing the robot to move wirelessly without physical connections.

4.1.2 Key Features

The ESP32 microcontroller offers several important features that make it suitable

for robotics applications.

- Built-in Bluetooth and Wi-Fi communication
- Dual-core processor with high processing speed
- Multiple GPIO pins for device interfacing
- Supports PWM for motor control
- Low power consumption
- High reliability and stability
- Compact size and easy programming

The ESP32 provides various communication interfaces such as:

- UART
- SPI
- I2C
- PWM

These interfaces enable easy integration with the motor driver module, sensors, and other electronic components used in the robot.

Another important feature of the ESP32 is its energy efficiency. It supports different power-saving modes such as:

- Deep sleep mode
- Light sleep mode
- Modem sleep mode

These modes help reduce power consumption in battery-operated systems like mobile robot

4.1.3 Working Principle

The ESP32 microcontroller operates by executing programmed instructions stored in its memory. In the Surveillance Bot system, it receives control commands from the mobile application through Bluetooth communication. The microcontroller processes these commands according to the programmed logic

and generates output signals that control the motor driver module.

For example:

- When the forward command is received, the ESP32 sends signals to rotate all motors forward
- When the backward command is received, the motors rotate in reverse direction
- When the left or right command is received, the motors rotate accordingly to change direction

The ESP32 continuously monitors incoming commands and controls the robot movement in real time. This ability to process data and control devices simultaneously makes the ESP32 a highly efficient controller for robotic systems. The ESP32 microcontroller offers several advantages over traditional microcontrollers. It combines high processing capability with built-in wireless communication, reducing the need for additional hardware. Its cost-effectiveness makes it suitable for both small-scale and large-scale robotics projects.

In conclusion, the ESP32 microcontroller is a versatile and powerful component that plays a vital role in the operation of the Surveillance Bot surveillance robot. Its combination of high performance, wireless connectivity, low power consumption, and ease of use makes it an ideal choice for mobile robotic applications.

ESP32 DEV. BOARD PINOUT

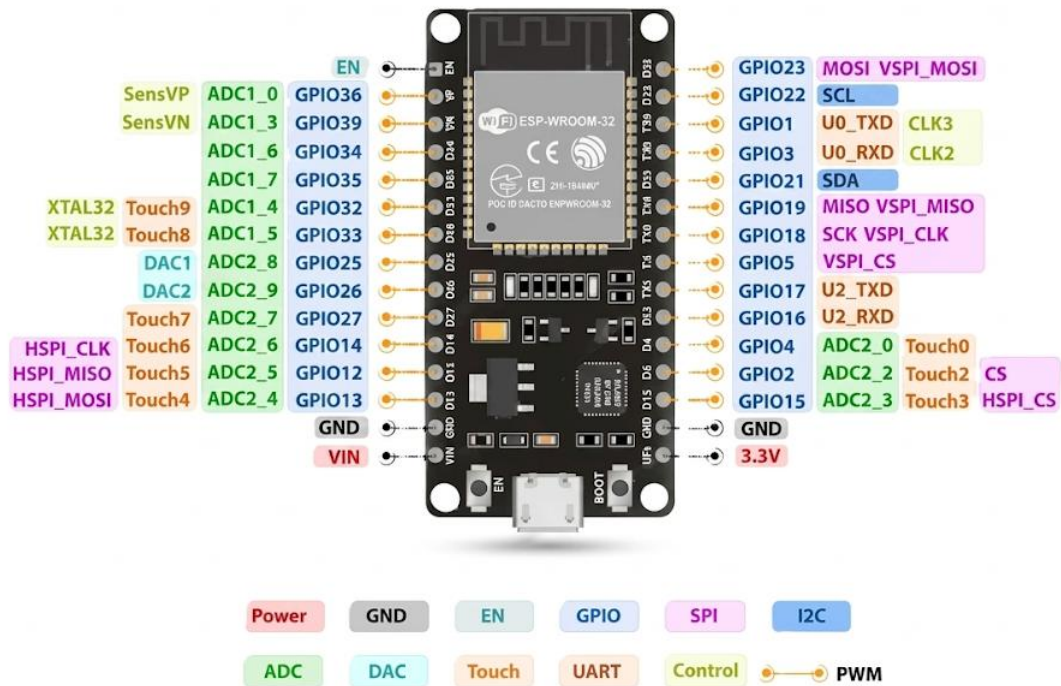


Figure 4.1 ESP32 DEV Microcontroller

4.2 POWER SUPPLY

A power supply is an electrical device that provides electric power to an electrical load. The main purpose of a power supply is to convert electrical energy from a source into the correct voltage and current required to operate electronic devices safely and efficiently. In the Surveillance Bot system, the power supply provides electrical energy to the ESP32 microcontroller, motor driver module, and DC motors. A stable power supply is essential to ensure reliable operation of the robot during movement. Power Supply in the Robot System The surveillance robot is powered by a rechargeable lithium-ion battery, which provides a compact and portable source of electrical energy. The battery supplies power to all components of the system, allowing the robot to operate without being connected to an external power source.

Since DC motors require higher current compared to the microcontroller, a motor driver module is used to distribute regulated power to the motors while protecting

the ESP32 from excessive current.

The use of a rechargeable battery makes the robot portable, lightweight, and suitable for remote monitoring applications.

Functions of Power Supply

The power supply performs several important functions:

- Converts electrical energy into usable voltage
- Supplies power to electronic components
- Maintains stable voltage levels
- Protects components from damage
- Ensures reliable system operation

Proper voltage regulation helps prevent sudden shutdown and improves the lifespan of electronic components.

Advantages of Rechargeable Battery

- Portable and lightweight
- Rechargeable and reusable
- Provides stable power output
- Suitable for mobile applications
- Environment friendly

In conclusion, the power supply plays a critical role in the operation of the surveillance robot by providing reliable and stable electrical energy to all components.



Figure 4.2 lithium-ion battery

4.3 MOTOR DRIVER MODULE (L298N)

A motor driver module is an electronic device used to control the speed and direction of motors. It acts as an interface between the microcontroller and the motors. In the Surveillance Bot system, the motor driver module is responsible for controlling the movement of the DC motors based on signals received from the ESP32.

The motor driver used in this project is typically the L298N motor driver, which can control multiple motors simultaneously.

Working Principle

The ESP32 microcontroller sends low-power digital signals to the motor driver module. The motor driver then converts these signals into high-power electrical signals required to operate the motors.

This process allows the motors to rotate in different directions depending on the input signals.

For example:

- Forward movement — motors rotate forward
- Backward movement — motors rotate backward
- Left turn — left motors stop, right motors rotate
- Right turn — right motors stop, left motors rotate

The motor driver ensures smooth and controlled motor operation.

Features of Motor Driver

- Controls motor speed and direction
- Supports bidirectional motor rotation
- Provides high current output
- Protects the circuit from overload
- Easy to interface with microcontrollers

Advantages

- Improves motor performance
- Ensures safe operation
- Reduces load on microcontroller
- Provides stable power supply

In conclusion, the motor driver module plays an essential role in controlling the movement of the robot and ensuring smooth operation.

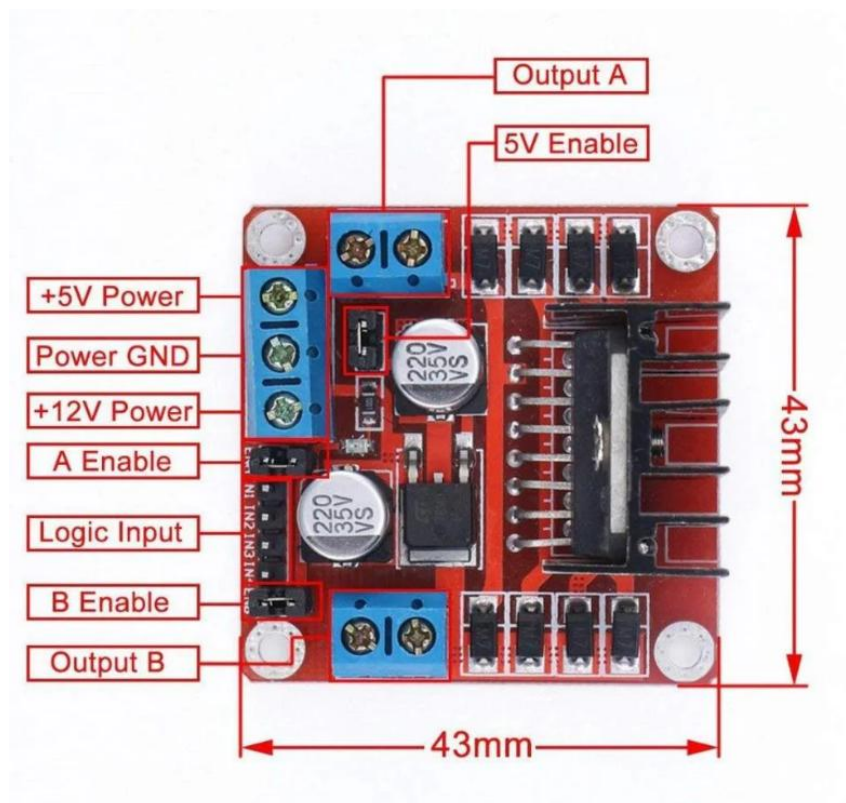


Figure 4.3 L298N Motor Driver

4.4 DC MOTOR

A DC motor is an electrical device that converts electrical energy into mechanical energy. It is widely used in robotics to provide movement and motion. In the Surveillance Bot project, four DC motors are used to drive the wheels of the robot. Each motor is connected to the motor driver module, which controls its rotation speed and direction. The motors rotate clockwise or counterclockwise depending on the control signals received from the microcontroller.

Working Principle

The DC motor operates on the principle of electromagnetic induction. When electrical current flows through the motor coil, a magnetic field is created, causing the motor shaft to rotate.

This rotation produces mechanical motion that moves the robot wheels.

Features of DC Motor

- Converts electrical energy into mechanical motion
- Provides continuous rotation
- Simple and reliable design
- Low cost and easy maintenance
- Suitable for robotic applications

Applications

- Robotics
- Automated vehicles
- Industrial machines
- Conveyor systems
- Home appliances

Advantages of DC Motors

- High efficiency
- Simple control mechanism
- Low maintenance
- Long operating life
- Reliable performance

In conclusion, the DC motor is a key component responsible for the movement and mobility of the surveillance robot.



Figure 4.4 DC motor

4.5 ROBOT CHASSIS

The robot chassis is the mechanical structure that supports all components of the robot. It provides a stable base for mounting motors, wheels, battery, and electronic circuits. The chassis ensures proper alignment and balance during movement.

In the Surveillance Bot system, a four-wheel drive chassis is used to provide stability and smooth movement on different surfaces.

Functions of Robot Chassis

- Supports all hardware components
- Maintains structural stability
- Protects internal components
- Enables smooth movement

Features of Robot Chassis

- Strong and durable structure
- Lightweight design
- Easy to assemble
- Provides stability and balance

Advantages

- Improves robot performance
- Reduces vibration
- Ensures safe operation
- Increases durability

In conclusion, the robot chassis plays a vital role in maintaining the structural strength and stability of the surveillance robot.

CHAPTER 5

SYSTEM ARCHITECTURE

5.1 OVERVIEW

The Surveillance Bot system is a mobile-controlled surveillance robot designed to perform remote monitoring and navigation tasks. Its main purpose is to provide wireless control of a robot using a smartphone to perform operations such as moving forward, backward, turning, and stopping. This system improves safety, efficiency, and convenience in surveillance applications such as home security, industrial monitoring, and restricted area inspection.

The system uses Bluetooth communication, embedded control, and DC motor actuation to convert user commands into mechanical movement of the robot. The mobile device acts as the input device, while the ESP32 microcontroller processes the commands and controls the movement of the robot through the motor driver and DC motors.

The architecture of the system is designed using a modular approach. Each component performs a specific function and communicates with other components to ensure smooth operation. This design improves system reliability, flexibility, and ease of maintenance.

The system operates wirelessly without requiring internet connectivity. This allows the robot to function in remote locations where internet service may not be available. The overall architecture ensures efficient communication between components and enables reliable operation of the surveillance robot.

5.2 SYSTEM ARCHITECTURE DESIGN

The system architecture design describes how different components of the robot interact to perform the required functions. The operation of the system is divided into four main stages:

- Input
- Processing
- Control
- Output

Initially, the system is powered using a rechargeable battery power supply. The battery provides electrical energy to the ESP32 microcontroller, motor driver module, and DC motors. Since DC motors require higher current for operation, a motor driver module is used to supply regulated voltage and protect the microcontroller from damage.

The process begins when the user sends a command using a mobile application installed on a smartphone. The command is transmitted through Bluetooth communication to the ESP32 microcontroller.

Once the command is received, the ESP32 processes the input signal according to the programmed instructions stored in its memory. The microcontroller then sends control signals to the motor driver module, which controls the speed and direction of the DC motors.

The system uses four DC motors to drive the wheels of the robot. Each motor rotates in a specific direction based on the command received from the microcontroller.

For example:

- Forward command — All motors rotate forward
- Backward command — All motors rotate backward
- Left command — Right motors rotate while left motors stop
- Right command — Left motors rotate while right motors stop
- Stop command — All motors stop rotating

The coordinated movement of the motors allows the robot to move smoothly in different directions. The system continuously receives commands from the mobile device, enabling real-time control of the robot.

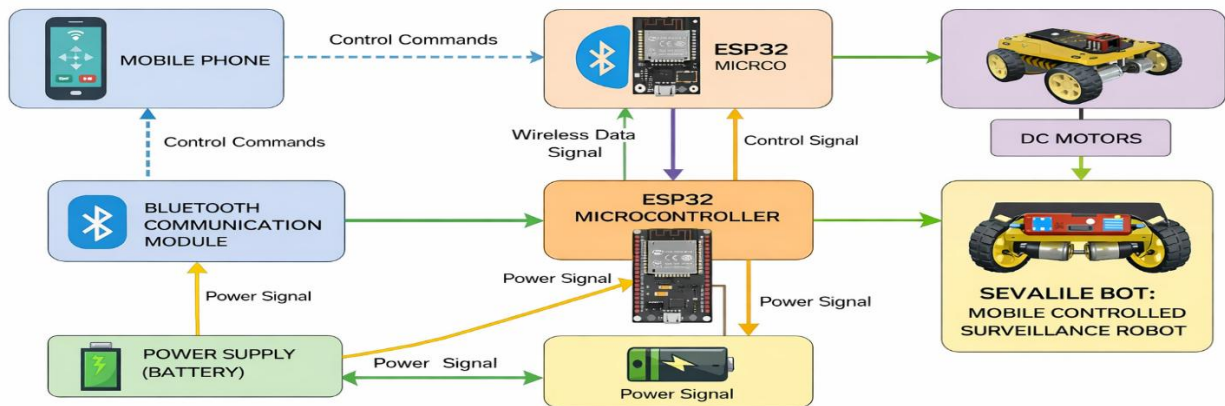


Figure 5.2 System Architecture

5.3 SYSTEM CHARACTERISTICS

1. Wireless Operation

The system allows users to control the robot wirelessly using a mobile device. Bluetooth communication eliminates the need for physical connections between the user and the robot.

This feature provides flexibility and convenience, especially in surveillance applications where remote control is required.

Design Consideration

Reliable Bluetooth connectivity ensures stable communication between the mobile device and the robot.

2. Real-Time Response

The system is capable of responding quickly to user commands. The ESP32 microcontroller processes incoming signals immediately and controls the motors without delay.

Real-time response improves control accuracy and ensures smooth robot movement.

Design Consideration

Efficient programming and fast communication reduce system response time.

3. Offline Functionality

The system operates without requiring internet connectivity. All commands are transmitted directly through Bluetooth communication.

This feature makes the system suitable for use in remote areas or locations without internet access.

Design Consideration

Local communication ensures reliable operation and fast response.

4. Easy Control Interface

The mobile application provides a simple and user-friendly interface for controlling the robot.

Users can easily understand and operate the system without technical knowledge.

Design Consideration

Clear button layout improves usability.

5. Safety and Reliability

The system includes safety features to protect both the hardware and the user.

Safety Measures

Power Regulation

The motor driver provides stable voltage to the motors.

Overload Protection

Prevents damage to motors and electronic components.

Stable Wiring

Ensures reliable communication between components.

These safety measures improve system performance and durability.

6. Scalability

The system is designed to support future expansion. Additional components can be added without changing the basic system design.

Examples of additional components:

- Camera module
- Obstacle detection sensor
- Temperature sensor
- GPS module

This flexibility makes the system suitable for advanced surveillance applications.

5.4 SYSTEM FLOW

The system flow describes the sequence of operations that occur during the functioning of the robot. It shows how commands move through different components of the system.

Step 1: Command Input

The user sends a command using a mobile application.

Examples of commands:

- Forward
- Backward
- Left
- Right
- Stop

Step 2: Signal Transmission

The mobile device transmits the command to the ESP32 microcontroller using Bluetooth communication.

Step 3: Command Processing

The ESP32 microcontroller receives the command and processes it according to the programmed instructions.

Step 4: Motor Control

The microcontroller sends control signals to the motor driver module.

The motor driver controls the speed and direction of the DC motors.

Step 5: Robot Movement

The DC motors rotate according to the control signals.

The robot moves in the required direction.

Step 6: Continuous Operation

The system continuously receives commands and performs operations in real time.

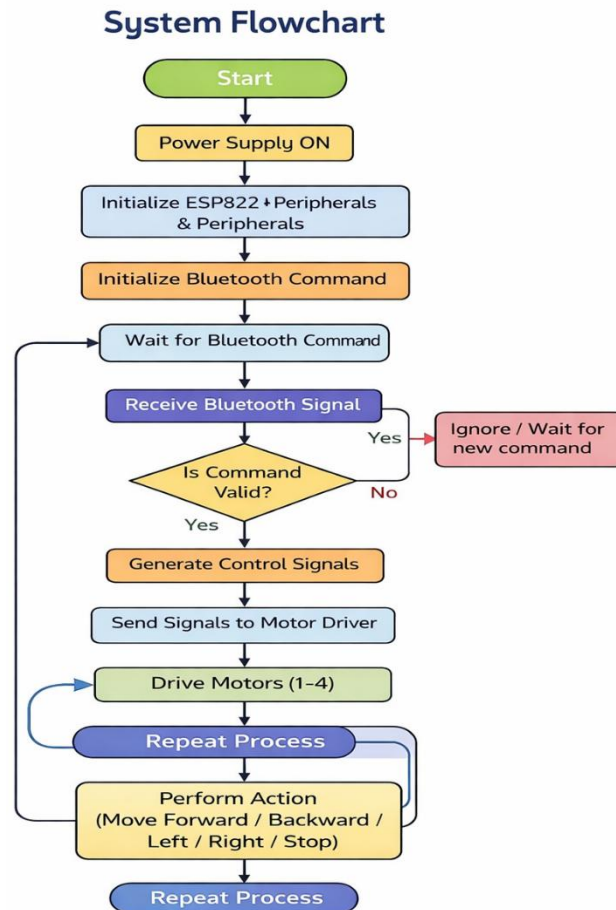


Figure 5.4 System Flow

CHAPTER 6

RESULT AND DISCUSSION

6.1 RESULT

The developed Surveillance Bot – Mobile Controlled Surveillance Robot was successfully designed, assembled, and tested to demonstrate reliable wireless control of a robotic vehicle using an ESP32 microcontroller and Bluetooth communication. The integration of the ESP32 controller, motor driver module, DC motors, Bluetooth communication system, and power supply resulted in a functional robot capable of performing basic movement operations such as forward, backward, left, right, and stop.

During testing, the system responded accurately to control commands sent from a mobile application. The Bluetooth communication module established a stable wireless connection between the smartphone and the robot, allowing real-time command transmission. The ESP32 microcontroller processed the received commands efficiently and generated appropriate control signals for the motor driver module. The motor driver successfully controlled the rotation of the DC motors based on the received signals. As a result, the robot demonstrated smooth and stable movement in all directions. The coordinated operation of the motors enabled precise navigation and improved manoeuvrability of the robot.

The power supply unit provided stable voltage to all components, ensuring uninterrupted system operation. The system operated continuously without overheating or power fluctuations during testing. The robot structure remained stable during repeated operations, confirming the mechanical strength and reliability of the design.

Performance testing was conducted in indoor environments to evaluate the response time and reliability of the system. The robot showed fast response to user commands with minimal delay. The communication range of the Bluetooth

connection was observed to be effective within approximately 10 meters, which is suitable for short-distance surveillance and monitoring applications.

Overall, the developed system demonstrated reliable operation, stable communication, and effective movement control, confirming the successful implementation of the mobile-controlled surveillance robot.

6.2 DISCUSSION

The discussion of results indicates that the proposed system successfully meets the objective of developing a mobile-controlled surveillance robot capable of performing remote monitoring and navigation tasks.

The use of an ESP32 microcontroller provides efficient processing capability, allowing the system to handle multiple motor control operations simultaneously. Compared to traditional wired control systems, the Bluetooth-based wireless control system offers greater flexibility and convenience.

The robot can be operated remotely without physical contact, improving safety and usability in surveillance applications. The simple control interface also makes the system easy to operate for users without technical knowledge.

The modular design of the system allows easy integration of additional components such as cameras, sensors, or GPS modules. This flexibility enables future expansion of the robot for advanced applications such as obstacle detection, remote surveillance, and environmental monitoring.

Despite the successful performance of the system, some limitations were identified during testing. The Bluetooth communication range is limited compared to Wi-Fi-based systems, which restricts long-distance operation. Additionally, the robot's performance depends on battery capacity, and extended usage may require frequent recharging.

These limitations provide opportunities for future improvements, such as:

- Using Wi-Fi communication for extended range

- Adding a rechargeable high-capacity battery
- Integrating obstacle detection sensors
- Installing a camera module for live video surveillance

Final picture of the project

Figure 6.2 *Figure 6.2 Final Prototype of Surveillance Bot – Mobile Controlled Surveillance Robot Using ESP32*

CHAPTER 7

CONCLUSION

The Surveillance Bot – Mobile Controlled Surveillance Robot using ESP32 project successfully demonstrates the design and implementation of a wireless robotic system capable of performing remote control operations through Bluetooth communication. The system integrates an ESP32 microcontroller, motor driver module, DC motors, Bluetooth communication module, and power supply to achieve efficient robot movement and control.

The developed prototype effectively converts commands from a mobile device into mechanical movement of the robot. The robot can move in different directions, including forward, backward, left, right, and stop, based on user instructions. The results confirm that the system operates reliably with fast response time and stable performance.

The ESP32 microcontroller plays a vital role in ensuring smooth processing and control of the DC motors. Its efficient signal generation enables accurate motor control, contributing to stable and precise robot movement. The power supply and motor driver components provide reliable electrical performance, ensuring continuous operation of the system.

One of the major achievements of this project is the development of a low-cost, user-friendly, and reliable surveillance robot. The system improves safety and convenience by allowing remote operation without direct human involvement.

This makes the robot suitable for applications such as:

- Home security
- Industrial monitoring
- Hazardous area inspection
- Remote surveillance

Although the system performs well under normal operating conditions, certain limitations were observed, such as limited Bluetooth range and battery

dependency. These limitations can be addressed through future enhancements, including Wi-Fi communication, camera integration, and advanced sensors. Overall, the project demonstrates the successful implementation of a practical and efficient mobile-controlled robotic system.

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